Memory Map

ELL7282 Assignment 2 · Dharun Thota (Entry No. 2026EET2401)

Address decode (8-bit data address, decoded from the top bits only)

Range	Region	Selected when	Read	Write
0x00–0x7F	RAM, 128 bytes	ADDR[7] = 0; RAM address = ADDR[6:0]	stored byte	stores byte
0x80–0xEF	reserved	ADDR[7] = 1 and ADDR[7:4] ≠ 1111	0x00	ignored
0xF0–0xFF	peripherals	ADDR[7:4] = 1111; register index = ADDR[3:0]	see below	see below

Peripheral registers

Addr	Name	Access	Reset	Bits and behaviour
0xF0	IN_STATUS	R	flags 0	[0] START (live pin), [1] DOOR_CLOSED (live pin), [3:2] MODE (live pins), [4] START_PEND (sticky; set by a 0→1 edge of START, cleared by ACK_START), [5] TRIP (safety latch), [7:6] = 0
0xF1	WATER_LEVEL	R	live	[7:0] unsigned level
0xF2	TEMP	R	live	[7:0] temperature reading (read only, no control loop)
0xF3	DRUM_RPM	R	live	[7:0] drum speed reading
0xF4	ACT	R/W	0x00	actuator request latch: [0] VALVE_IN, [1] HEATER, [2] MOTOR_EN (request), [3] MOTOR_DIR, [4] PUMP_OUT, [5] DOOR_LOCK, [6] BUZZER, [7] reads 0. Reads return the latch, not the gated motor output
0xF5	DISPLAY	R/W	0x00	[7:0] status code: 01 idle, 02 fill, 03 wash, 04 drain, 05 spin, 0D done
0xF6	SYS_CTRL	W	–	[0] ACK_START strobe: clears START_PEND, and rearms TRIP when START_PEND = 1, DOOR_CLOSED = 1 and MOTOR_EN request = 0. Reads 0
0xF7	–	–	–	unimplemented: reads 0, writes ignored
0xF8	TIMER_COUNT	R	0x00	[7:0] current count
0xF9	TIMER_LIMIT	R/W	0x00	[7:0] compare value; 0 disables the comparison
0xFA	TIMER_CTRL	R/W	0x00	[0] EN (reads back as written); [1] CLR strobe, reads 0; [7:2] ignored, read 0
0xFB	TIMER_STATUS	R	0x00	[0] MATCH (sticky), [1] EN mirror, [7:2] = 0
0xFC–0xFF	–	–	–	unimplemented: reads 0, writes ignored

Rules

- A write to a read-only, reserved or unimplemented address changes nothing; reads have no side effects.
- A register is written once, in MEM, at the edge that ends the cycle of a *valid* store: `P_WE` = EXMEM_V ∧ EXMEM_isST ∧ (ADDR[7:4] = 1111). Bubbles and flushed instructions never reach a peripheral.
- Inputs are read while a valid LD is in MEM and captured at that edge; tests hold sensor pins steady through it. Every access completes in one cycle (no READY protocol).
- RESET clears ACT, DISPLAY, the timer (EN, COUNT, LIMIT, MATCH), START_PEND, the START edge detector and TRIP.

Timer

All timer registers use the global rising-edge clock with no stall or halt term, so the timer counts through stalls and after HALT. With values from before the edge, CLR = (store to 0xFA with bit 1 = 1):

$$\text{NEW} = (\text{COUNT} + 1) \bmod 256, \qquad \text{HIT} = \text{EN} \wedge \neg\text{CLR} \wedge (\text{LIMIT} \neq 0) \wedge (\text{NEW} = \text{LIMIT})$$

$$\text{COUNT}' = \text{CLR}\ ?\ 0 : (\text{EN}\ ?\ \text{NEW} : \text{COUNT}), \qquad \text{MATCH}' = \neg\text{CLR} \wedge (\text{MATCH} \vee \text{HIT})$$

$$\text{EN}' = \text{bit 0 of a store to 0xFA, else EN}, \qquad \text{LIMIT}' = \text{data of a store to 0xF9, else LIMIT}$$

So a clear beats the increment and the compare at the same edge, writing LIMIT alone never sets MATCH, LIMIT = 0 never matches, and COUNT wraps 255 → 0 while a set MATCH stays set. Example: LIMIT = 3 and a store of 0x03 (EN and CLR) to TIMER_CTRL at edge E give COUNT = 0, 1, 2, 3, 4 and MATCH = 0, 0, 0, 1, 1 after edges E to E+4; MATCH then stays 1 until the next clear. Software rearms an interval by writing LIMIT and then 0x03 to TIMER_CTRL (lock delay, each wash half-period, spin).